

Manual Version 0

1. Project set-up

1.1 Creating/Loading a project

The project will be set-up upon creating an instance of one of the **Manager** classes (see section 2), e.g.:

```
swam = MASW2DManager(<prjdir>,path2raw=<optinal path2raw>path2geom=<optinal  
path2geom>, settings=settings)
```

The project set-up consists of the following steps:

1. Creating the project directory structue (if it does not exist)
2. Copying (and by default renaming) raw data into the project directly (if it does not exist there and a path to the raw data was provided)
3. Copying or attempting to create the geometry file (if it does not exist in project directory and 1. a path to the geom file was provided and/or 2. source/receiver is contained in seismic data)
4. Loading the data
5. Creating the database

If a project already exists, upon creating an instance of the **Manager** and setting the correct project path, the project (database) will be loaded:

```
swam = MASW2DManager(<prjdir>)
```

1.1.1 Project directory

```
|prjdir  
|- 01_data/  
    - raw/  
|- 02_geom/  
|- 03_proc/  
|- 04_figs/
```

1.1.2 Geometry file

The geometry file is a csv file introduced and used in the open-source python library formikoj for seismic refraction tomography data processing and modelling (Steiner & Flores Orozco, 2023) that stores an abstract representation of the survey layout. In our toolbox it is passed the methods to provide the information about the survey layout. We slightly adapt the format to store the information for repeated shots at the same location. The file can be created manually using any (spreadsheet) editor or from the seismic data directly if the source and receiver coordinates are contained within the seismic raw data files. The file contains (without a header) in sorted order column-wise: (1) the local station coordinates (x-, y-, and z-coordinates), (2) a flag to indicate wether this location is a geophone or source, (3) the shot file number, (4) the first geophone and (5) the number of active geophones (see Table 1). Height and 3D information is not taken into consideration during the analysis or visualisations in the current version of the surface wave

processing toolbox. An example is given in Table 2. For further information check out Steiner & Flores Orozco (2023).

Table 1: Description of the columns of the geometry file

Column	Content	Data type	Description
1	x-coordinate	float	Station x coordinate, e.g., given in (m)
2	y-coordinate	float	Station y coordinate, e.g., given in (m)
3	z-coordinate	float	Station z coordinate, e.g., given in (m)
4	Geophone flag	int	1 if a geophone was deployed at station, 0 otherwise
5	Shot file number (s) separated by ;	str	The numerical part of the file name, e.g., 1001, if a shot was repeated and should be analysed, -1 otherwise
6	First geophone	str	First active geophone
7	Number of geophones	str	Number of active geophones (-1 if no shot was conducted at the station)

Table 2: Example of the geometry file structure

x (m)	y (m)	z (m)	Geo	Shot	1 st Geo	# Geo
0.0	0.0	0.0	1	1001;1002;1003;1004	1;1;1;1	48;48;48;48
2.0	0.0	0.0	1	1005; 1006	1;1	48;48
...	..	..	..	..	..	..
20.0	0.0	0.0	1	-1	1	-1
..	..	..	..	..	..	..
24	0.0	0.0	0	1023; 1024; 1025; 1026	10;10;10;10	48;48;48;48

References:

Steiner, M., & Orozco, A. F. (2023). formikoj: A flexible library for data management and processing in geophysics—Application for seismic refraction data. *Computers & Geosciences*, 176, 105339.

2. Manager modules

BaseManager

Base manager class to:

- create project directory
- load seismic raw data
- store/access data in database
- apply functions from SeismicStream, DispersionCurve and CombineCurves classes to a single data set
 - preprocessing (e.g., frequency filtering, trimming)
 - wavefield transformation
 - dispersion curve picking
 - processing of the dispersion curves
 - plotting
- export to file: dispersion curves

MASW2DManager

Class to process seismic data to run MASW workflows

- run base functions on a all data sets or a selection
- windowing
- export to file: dispersion curves

CDCManager

Class to perform the dispersion curve combination

- curve combination
- export to file: dispersion curves

Tomo2DManager

Class to process seismic data to run Tomographic-like approach workflows

- preprocess stream
- computation of phase differences
- run the inversion

3. Database

A database is created when using the manager module, in which data from different processing steps is stored, that can be accessed at later points.

Tables currently stored in the data base are:

- geom
- shots
- recs
- settings
- amps
- par
- pid
- FV
- freq
- pd
- curve

1. Tables defining the survey geometry

Table 3: Table geom

Column	Name	Content	Data type	Description
1	station_id	Station id	int	Station id
2	x	x-coordinate	float	Station x coordinate, e.g., given in (m)
3	y	y-coordinate	float	Station y coordinate, e.g., given in (m)
4	z	z-coordinate	float	Station z coordinate, e.g., given in (m)
5	geophone	Geophone flag	int	1 if a geophone was deployed at station, 0 otherwise

6	shots	Shot file number(s)	str	The numerical part of the file name, e.g., 1001, if a shot was conducted at this station, -1 otherwise
7	first_geo	First geophone	int	First active geophone (currently not used, but in later versions relevant for roll-along data)
8	num_geo	Number of geophones	int	Number of active geophones (-1 if no shot was conducted at the station)

Table 4: Table shots

Column Name	Content	Data type	Description
1	station_id	Station id	int
2	sin	Source location id	int
3	rep	Shot repetition id	int

Table 5: Table recs

Column Name	Content	Data type	Description
1	station_id	Station id	int
2	rin	Receiver id	int

2. Tables containing seismic data

Table 6: Table amps

Column Name	Content	Data type	Description
1	procset	Processing set label	str
2	sin	Source location id	int
3	rep	Shot repetition id	int
4	wid	Window id	int
5	rin 1	Processed trace data	float
...	...	...	...
n	rin n	Processed trace data	float

Table 7: Table par

Column Name	Content	Data type	Description
1	procset	Processing set label	str
2	sin	Source location id	int
3	rep	Shot repetition id	int
4	wid	Window id	int
5	npts	Number of samples	int
6	dt	sampling interval in s	float
7	delay	pre trigger in s	float
8	sampling_rate	sampling rate	float

Table 8: Table pid

Column Name	Content	Data type	Description
1	procset	Processing set label	str

2	sin	Source location id	int
3	rep	Shot repetition id	int
4	wid	Window id	int
5	rin	Receiver id	int

Table 9: Table FV

Column Name		Content	Data type	Description
1	procset	Processing set label	str	Method to calculate the dispersion image
2	sin	Source location id	int	
3	rep	Shot repetition id	int	
4	wid	Window id	int	
5	method	method	str	
6	velocity	Testing phase velocity	float	
7	wavenumber	wavenumber	float	
8	f1	Dispersive energy at frequency 1	complex	
...	...			
n	fn	Dispersive energy at frequency n	complex	

Table 10: Table freq

Column Name		Content	Data type	Description
1	procset	Processing set label	str	Method to calculate the dispersion image
2	sin	Source location id	int	
3	rep	Shot repetition id	int	
4	wid	Window id	int	
5	method	method	str	
6	frequency	Frequency	float	Frequency range used to compute the dispersive energy

Table 11: Table pd

Column Name		Content	Data type	Description
1	procset	Processing set label	str	Method to create aggregate
2	sin	Source location id	int	
3	rep	Shot repetition id	int	
4	wid	Window id	int	
5	calc	Aggregate operator	str	
6	fids	Frequency ids	int	Frequency range used to compute the dispersive energy
7	frequency	Frequency	float	
8	pd1	Phase difference	float	Phase difference computed between rin1 and rin2
...				
n	pdn	Phase difference	float	Phase difference computed between rin n-1 and rinn

Table 12: Table curve

Column	Name	Content	Data type	Description
1	procset	Processing set label	str	
2	sin	Source location id	int	
3	rep	Shot repetition id	int	
4	wid	Window id	int	
5	xmid	Receiver midpoint	float	Assumed location of the dispersion curve
6	method	method	str	Method to obtain the dispersion curve
7	dc_mode	Mode of the dispersion curve	int	
8	frequency	Frequency	float	
9	velocity	Phase velocity	float	
10	error	Dispersion curve absolute error	float	